CAI6734 Weekly Report 1

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1 Project Objective

The goal of this project is to train a search agent capable of navigating structured EHR data. This week's scope was to reproduce the methodology presented in the LA-CDM paper [1]. The goal of this task was to improve our understanding of how a pre-trained model can be adapted to support clinical decision-making.

The paper fine-tuned a pre-trained Qwen model using Low-Rank Adaptation (LoRA), a parameter-efficient method that freezes the original model weights and trains a smaller set of adapter parameters. These adapters were optimized using supervised fine-tuning and Group Relative Policy Optimization (GRPO) to improve hypothesis generation, confidence calibration, and diagnostic decision-making.

The model was trained and evaluated on a cohort of 2,400 MIMIC-IV admissions across four acute abdominal conditions: appendicitis, cholecystitis, diverticulitis, and pancreatitis. The cohort was divided into 1,920 training cases, 240 validation cases, and 240 test cases.

2 Accomplishments

I obtained credentialed access to MIMIC-IV and used version 2.2 of the dataset on HiPerGator to reproduce the study's methodology. I used Meta-Llama-3-8B-Instruct in place of Qwen because Chinese-developed models were not permitted on HiPerGator. Since the paper's codebase was open source, I was able to clone it and save a copy on my personal account.

I modified the original LA-CDM codebase to support Llama and the HiPerGator environment. These changes included replacing Qwen-specific model paths, resolving dependency conflicts, adding Slurm job scripts, adapting the output parser to Llama's response format, reducing GPU memory usage, and adding validation scripts for both the cohort and prepared datasets. I also updated the evaluation pipeline to report diagnostic accuracy, calibration, test usage, diagnostic cost, and disease-specific performance.

I extracted a cohort from MIMIC-IV and MIMIC-IV-Note version 2.2 using the same four abdominal conditions studied in LA-CDM. The final validated cohort contained 2,400 admissions: 957 appendicitis cases, 648 cholecystitis cases, 257 diverticulitis cases, and 538 pancreatitis cases. I divided the cohort into 1,920 training cases, 240 validation cases, and 240 test cases. I then generated Llama-based patient-history summaries and validated the resulting files for missing summaries, duplicate admissions, malformed fields, diagnosis leakage, and overlap between dataset splits.

Finally, I evaluated the untrained Llama model on all 240 test cases to establish a zero-shot baseline. The model achieved 8.75% overall accuracy and produced no valid final diagnosis for 87.92% of cases. However, it achieved 72.41% accuracy among the cases for which it completed a valid diagnosis. This suggests that the model's main zero-shot limitation was following and

completing the required interaction protocol rather than identifying the correct condition. I also validated the LoRA training pipeline through short training, checkpoint, adapter-loading, and production-scale memory tests in preparation for the full training run.

The full 300-step LoRA training run is currently in progress. This run includes 100 Decision Agent reinforcement- learning steps, 100 Hypothesis Agent supervised fine-tuning steps, and 100 confidence-calibration reinforcement- learning steps. The run is expected to take several hours.

3 Challenges

One early challenge was waiting for credentialed access to the MIMIC-IV data before I could begin cohort extraction. I also had to replace the paper's Qwen model with Llama because Chinese-developed models are not permitted on HiPerGator. This required changes to model loading, prompting, output parsing, and evaluation behavior.

The original project also had Python and package-version conflicts that prevented its requirements from being installed directly. I created a compatible environment and moved it to Blue storage after reaching my home-directory quota. GPU memory was another recurring challenge. Several runs failed with out-of-memory errors, so I separated model workloads across two GPUs and added microbatching and chunked loss calculations to reduce peak memory use.

4 Next Steps

After training is complete, I will evaluate the trained LoRA adapter on the same 240 held-out test cases used for the zero-shot baseline. I will compare the trained and untrained models using overall diagnostic accuracy, valid diagnosis rate, disease-specific precision and recall, macro F1 score, calibration error, number of requested tests, and estimated diagnostic cost. This comparison will show whether training improves both diagnostic performance and compliance with the required clinical decision-making protocol.

I will then run the planned ablation experiments, including a Decision-Agent-only model and a version trained without the diagnostic test-cost reward. These experiments will help determine the contribution of the Hypothesis Agent, confidence-calibration objective, and cost-sensitive reward to the final model's performance.

I will also use Synthea [2] to generate a synthetic OMOP dataset containing approximately 5,000 patients for my teammates to begin using. Our group plans to pre-train an agent on OMOP data because OMOP is intended to provide a standardized, general-purpose data model across different healthcare systems.

Finally, I will take a deeper look at the paper's code to understand how its model-training pipeline works. I will use that understanding to begin designing experiments with the synthetic OMOP data that are currently available.

5 AI Disclaimer

I used GPT-5.6 Sol to guide me throughout this process and assist with writing this report. I also used it to summarize the LA-CDM paper. I reviewed and revised the resulting material and take responsibility for the final content.

References

[1] David Bani-Harouni, Chantal Pellegrini, Ege Özsoy, Nassir Navab, and Matthias Keicher. Language agents for hypothesis-driven clinical decision making with reinforcement learning. In *International Conference on Learning Representations*, 2026. https://arxiv.org/abs/2506.13474.

[2] Synthetic Health. Synthea: Synthetic patient population simulator. GitHub repository, https://github.com/synthetichealth/synthea, accessed September 29, 2026.